

Visual Complexity in VR: Implications for Cognitive Load

Maximilian Rettinger *

Xuefei Jiang

Jiaxin Yang

Gerhard Rigoll

Technical University of Munich

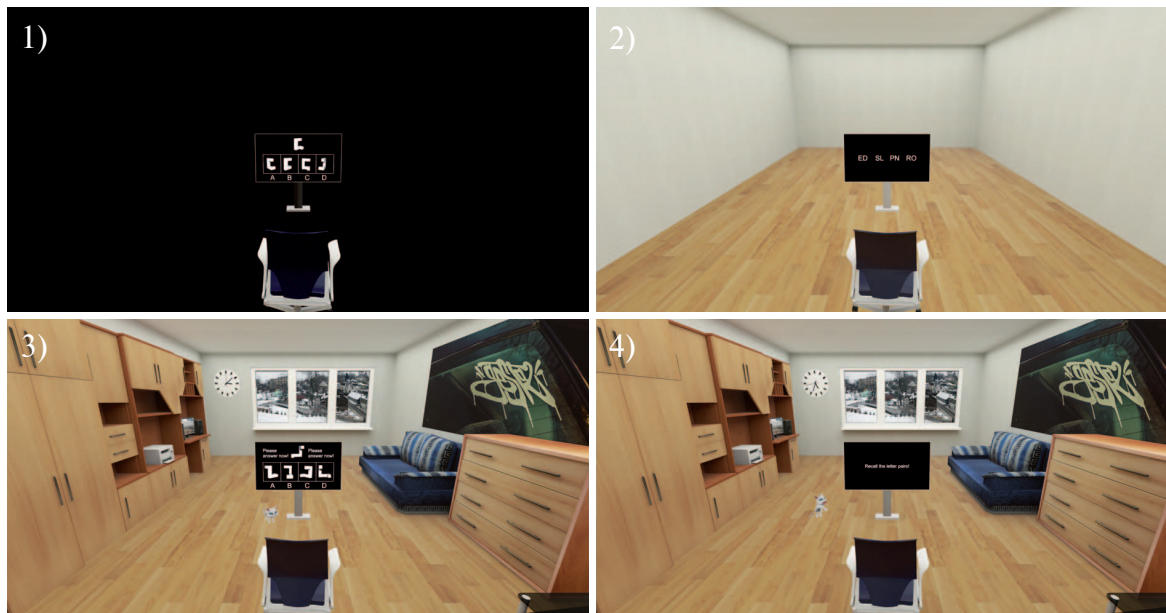


Figure 1: The compared virtual environments with different complexity levels: 1) *No-Room*, 2) *Empty-Room*, 3) *Filled-Room*, and 4) *Animated-Room*. The virtual monitor indicates the letter recall and mental rotation task.

ABSTRACT

Given the potential of Virtual Reality for education and training, it is essential to implement it as effectively as possible. For this reason, we investigate the impact of virtual environments with different complexity levels on the cognitive load. We compare the methods 1) *No-Room*, 2) *Empty-Room*, 3) *Filled-Room*, and 4) *Animated-Room*. In a within-subjects study ($n = 40$), participants completed a letter recall and a mental rotation task to assess the cognitive load. The results show significant differences in the letter recall task, indicating that the cognitive load is also lower in the rooms with less complexity.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models; Applied computing—Education—Interactive learning;

1 INTRODUCTION

One of the most promising application areas for Virtual Reality (VR) is the field of education and training. Publications already demonstrate the effectiveness [4, 5] of this technology, but it is still being determined how such an application should be implemented to be most effective for users. For successful training outcomes, the cognitive load, i.e., the mental effort required during learning

or performing a task in working memory, should be reduced to a minimum during such training.

Studies have already shown that cognitive load [8] is affected by the visual representation as a self-avatar [7], the avatar type [3], as well as a bystander [6]. Jung *et al.* investigated the influence of the design of the spatial environment on cognitive load [2]. This involved comparing the following environments: 1) Classroom, 2) Music room, 3) Science lab, and 4) Sports field. The results indicate that the design could affect the cognitive load. However, the complexity of the visual information mostly stayed the same in this study. Therefore, we compare the following methods in a within-subjects study ($n = 40$): 1) *No-Room*, 2) *Empty-Room*, 3) *Filled-Room*, and 4) *Animated-Room*. Our study is designed to assess the effects of virtual environments with different complexity levels on cognitive load.

2 METHODOLOGY

We assess the cognitive differences between the five conditions using letter recall and mental rotation tasks. We divide these two tasks into five sections following the experimental protocol of previous works [6, 7]. The order and tasks of these sections are as follows: 1) the user memorizes four letter pairs from the letter recall task (15 seconds), 2) five geometric figures are displayed in the mental rotation task; one is the reference object to which two identical but rotated objects must be assigned (15 seconds), 3) the user answers the mental rotation task (10 seconds), 4) the user recalls the four letter pairs of the letter recall task (10 seconds), and 5) pause until the following task sequence starts (5 seconds). The study procedure contains 27 sequences with different letter pairs and geometric figures. Three of these are used in a test scenario to familiarize the user with the procedure, and the

*e-mail: maximilian.rettinger@tum.de

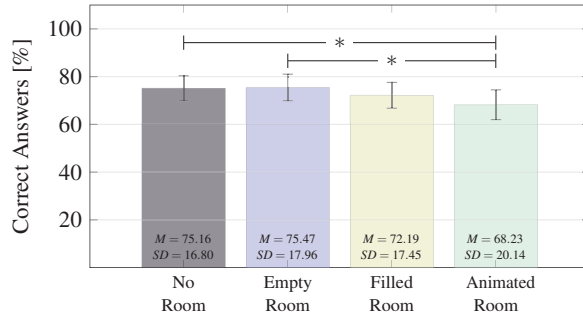


Figure 2: Correctly remembered letter pairs with 95% confidence intervals.

remaining six are distributed across the four conditions. For each condition, we display the tests on a virtual monitor (see Figure 1) while the user sits approximately 2 meters away on a swivel chair.

As depicted in Figure 1, the virtual environments differ in complexity. In *No-Room*, the user only sees the virtual screen for the tasks and a virtual chair on which he is also sitting in reality; *Empty-Room* extends the previous room with walls and light; *Filled-Room* is the extension of *Empty-Room* containing static objects such as furniture, a cat, and a window, as illustrated in Figure 1; and *Animated-Room* is based on the *Filled-Room* in which certain elements such as the cat, the clock, the world outside the window or the TV content are animated. The most striking feature is that the cat follows a circular path until it reaches the starting point and jumps up, after which the sequence is repeated. Overall, we designed these animations to correspond to a natural situation to avoid intrusive distractions for the user.

For the study, we recruited 40 participants (26 male, 14 female) aged between 19 and 33 ($M = 24.85$, $SD = 2.62$). All were students and completed the following procedure: 1) consent form and demographic questionnaire, 2) introduction to the hard- and software, 3) explanation of the procedure for the letter recall and mental rotation tasks using printed examples, 4) tutorial scene, and 5) main scenarios with the four conditions. The order of these conditions is equally distributed across the participants with the Latin square. We use the HTC Vive Pro Eye head-mounted display as the apparatus and implement the software using Unity3D.

3 RESULTS

We used one-way repeated measures analysis of variance (ANOVA) with $\alpha = 0.05$ and post-hoc paired t-tests with Bonferroni correction for statistical data analysis.

Letter Recall Participants correctly recalled letters are illustrated in Figure 2. The ANOVA shows a significant difference ($F(3, 117) = 3.763$, $p = 0.013$, $\eta_p^2 = 0.088$) between the four conditions. The pairwise comparison indicates significant differences between the *No-Room* and the *Animated-Room* ($p = 0.033$) as well as the *Empty-Room* and *Animated-Room* ($p = 0.023$) conditions. Participants performed best with the *No-Room* ($M = 75.16$, $SD = 16.80$) and *Empty-Room* ($M = 75.47$, $SD = 17.96$) conditions, followed by the *Filled-Room* ($M = 72.19$, $SD = 17.45$) and *Animated-Room* ($M = 68.23$, $SD = 20.14$) conditions.

Mental Rotation The statistical analysis of the correctly solved mental rotation tasks reveals no significant main effect ($F(3, 117) = 0.839$, $p = 0.475$, $\eta_p^2 = 0.021$). As depicted in Figure 3, the participants correctly completed most mental rotation tasks in the *Filled-Room* ($M = 93.75$, $SD = 7.25$) condition, followed by the *Empty-Room* ($M = 93.13$, $SD = 11.62$), *No-Room* ($M = 92.71$, $SD = 12.54$), and *Animated-Room* ($M = 90.42$, $SD = 14.56$) conditions.

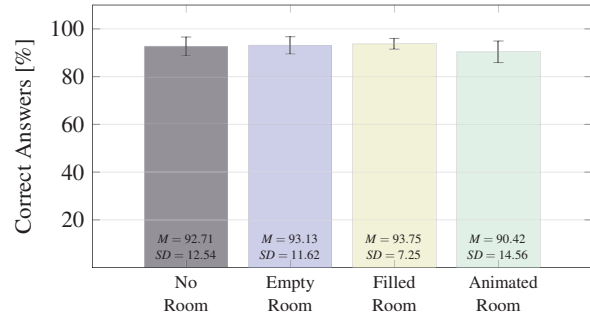


Figure 3: Correctly solved mental rotation tasks with 95% confidence intervals.

4 DISCUSSION AND CONCLUSION

This paper investigated the effect of spatial environments with different levels of complexity on the user's cognitive load. The results of a study ($n = 40$) showed that the performance in the letter recall task differed significantly between the conditions. The statistical analysis revealed no effect for the mental rotation task, i.e., the participants could solve these tasks reasonably well. However, the results of both tests indicate that the cognitive load is higher when users complete the tasks in an animated room. Our findings are consistent with the theory of extrinsic cognitive load [8].

Although the results indicate that the *No-Room* and *Empty-Room* conditions are the most suitable, it should be considered that the success of VR training can also depend on the context-dependent memory [1]. In other words, the user should complete the training in the same place where the knowledge will be recalled. Therefore, we recommend avoiding unnecessary information that does not affect the training content or content-dependent memory to design effective VR training.

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